

- 1 Suppose $y = -1 - 2x$.

When the value of x increases by 5, the value of y decreases by

- A 1 B 2
C 5 D 10 (1 mark)

(Q6, 2020 HSC) Medium

- 2 A plumber charges a call-out fee of \$90 as well as \$2 per minute while working.

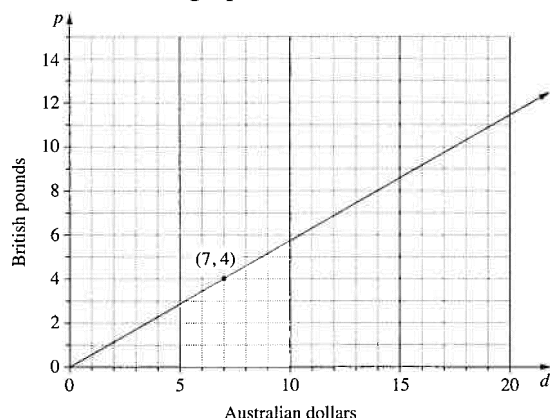
Suppose the plumber works for t hours.

Which equation expresses the amount the plumber charges (\$ C) as a function of time (t hours)?

- A $C = 2 + 90t$ B $C = 90 + 2t$
C $C = 120 + 90t$ D $C = 90 + 120t$ (1 mark)

(Q10, 2020 HSC) Medium

- 3 The relationship between British pounds (p) and Australian dollars (d) on a particular day is shown in the graph.



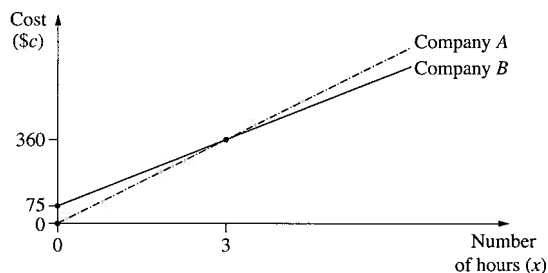
- a Write the direct variation equation relating British pounds to Australian dollars in the form $p = md$. Leave m as a fraction. (1 mark) Medium
- b The relationship between Japanese yen (y) and Australian dollars (d) on the same day is given by the equation $y = 76d$.

Convert 93 100 Japanese yen to British pounds. (2 marks) Medium

(Q34, 2019 HSC)

- 4 The graph displays the cost (\$ c) charged by two companies for the hire of a minibus for x hours.

Both companies charge \$360 for the hire of a minibus for 3 hours.



- i What is the hourly rate charged by Company A? (1 mark) Easy

- ii Company B charges an initial booking fee of \$75.

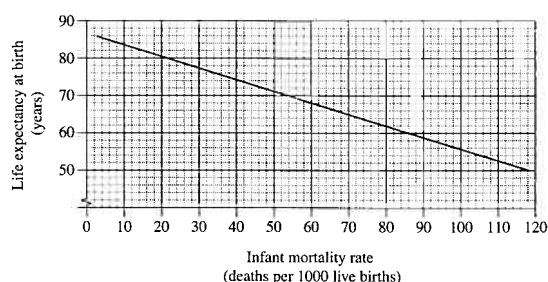
Write a formula, in the form of $c = mx + b$, for the cost of hiring a minibus from Company B for x hours. (2 marks) Easy

- iii A minibus is hired for 5 hours from Company B.

Calculate how much cheaper this is than hiring from Company A. (2 marks) Easy

(Q27d, 2018 HSC)

- 5 The graph shows the relationship between infant mortality rate (deaths per 1000 live births) and life expectancy at birth (in years) for different countries.



What is the life expectancy at birth in a country which has an infant mortality rate of 60?

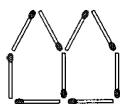
- A 68 years B 69 years
C 86 years D 88 years (1 mark)

(Q3, 2017 HSC) Easy

- 6 A pentagon is created using matches.



By adding more matches, a row of two pentagons is formed.



Continuing to add matches, a row of three pentagons can be formed.

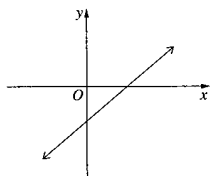


Continuing this pattern, what is the maximum number of complete pentagons that can be formed if 100 matches in total are available?

- A 25 B 24
C 21 D 20 (1 mark)

(Q20, 2017 HSC) Medium

- 7 The graph shows a line which has an equation in the form $y = mx + b$.

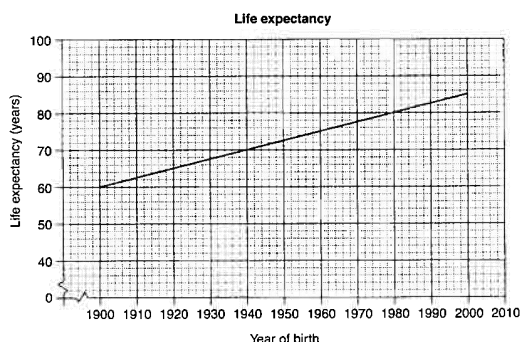


Which of the following statements is true?

- A m is positive and b is negative
B m is negative and b is positive
C m and b are both positive
D m and b are both negative (1 mark)

(Q14, 2016 HSC) Easy

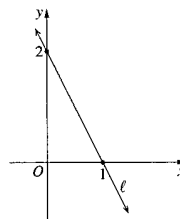
- 8 The graph shows the life expectancy of people born between 1900 and 2000.



- i According to the graph, what is the life expectancy of a person born in 1932? (1 mark) Easy
- ii With reference to the value of the gradient, explain the meaning of the gradient in this context. (2 marks) Easy

(29e, 2016 HSC)

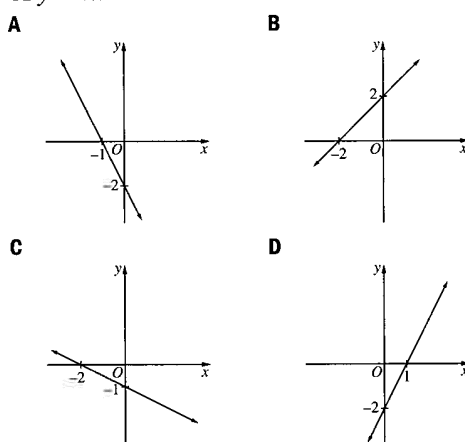
- 9 What is the equation of the line ℓ ?



- A $y = -2x + 2$ B $y = 2x + 2$
C $y = -\frac{x}{2} + 2$ D $y = \frac{x}{2} + 2$ (1 mark)

(Q13, 2015 HSC) Medium

- 10 Which of the following is the graph of $y = 2x - 2$?



(1 mark)

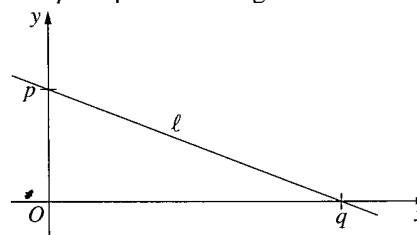
(Q7, 2014 HSC) Medium

- 11 The weight of an object on the moon varies directly with its weight on Earth. An astronaut who weighs 84 kg on Earth weighs only 14 kg on the moon.

A lunar landing craft weighs 2449 kg when on the moon. Calculate the weight of this landing craft when on Earth. (2 marks)

(Q26f, 2014 HSC) Easy

- 12 The line ℓ has intercepts p and q , where p and q are positive integers.



What is the gradient of line ℓ ?

- A $-\frac{p}{q}$ B $-\frac{q}{p}$
C $\frac{p}{q}$ D $\frac{q}{p}$ (1 mark)

(Q5, 2012 HSC) Medium

- 13 Dots were used to create a pattern. The first three shapes in the pattern are shown.



Shape 1

Shape 2

Shape 3

The number of dots used in each shape is recorded in the table.

Shape (S)	1	2	3
Number of dots (N)	6	8	10

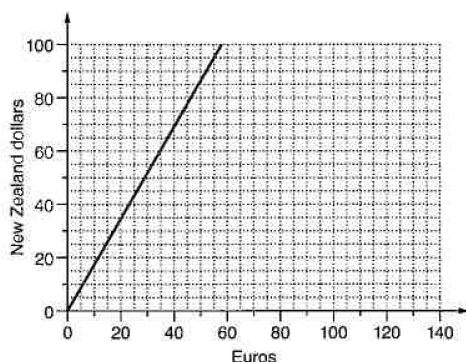
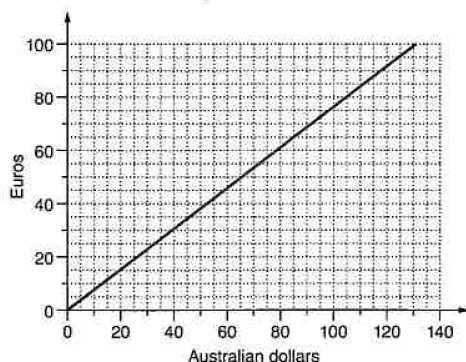
How many dots would be required for Shape 156?

- A 316 B 520
C 624 D 936

(1 mark)

(Q8, 2012 HSC) **Medium**

- 14 Conversion graphs can be used to convert from one currency to another.



Sarah converted 60 Australian dollars into Euros. She then converted all of these Euros into New Zealand dollars.

How much money, in New Zealand dollars, should Sarah have?

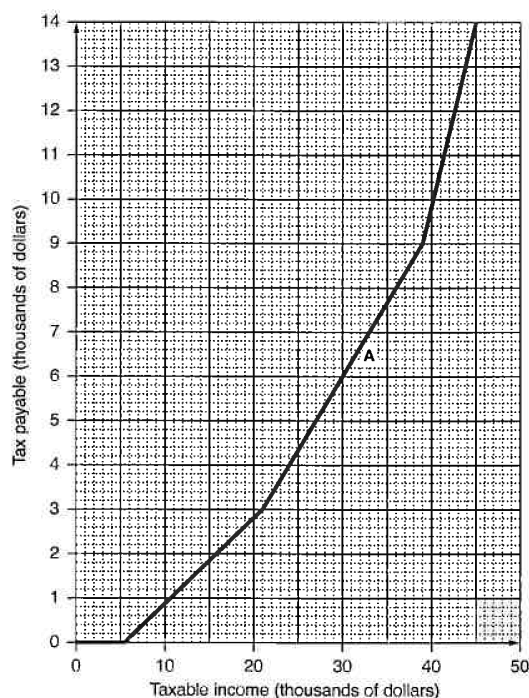
- A \$26 B \$45
C \$78 D \$135

(1 mark)

(Q13, 2012 HSC) **Easy**

- 15 The graph shows tax payable against taxable income, in thousands of dollars.

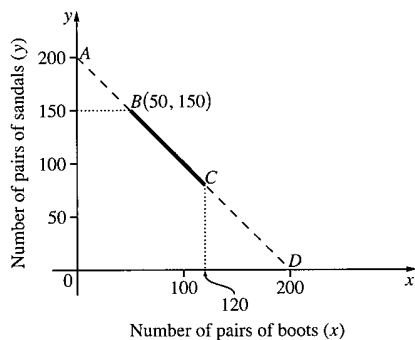
- Use the graph to find the tax payable on a taxable income of \$21 000. (1 mark) **Easy**
- Use suitable points from the graph to show that the gradient of the section of the graph marked A is $\frac{1}{3}$. (1 mark) **Medium**
- How much of each dollar earned between \$21 000 and \$39 000 is payable in tax? (1 mark) **Medium**
- Write an equation that could be used to calculate the tax payable, T , in terms of the taxable income, I , for taxable incomes between \$21 000 and \$39 000. (2 marks) **Hard**



(Q27c, 2010 HSC)

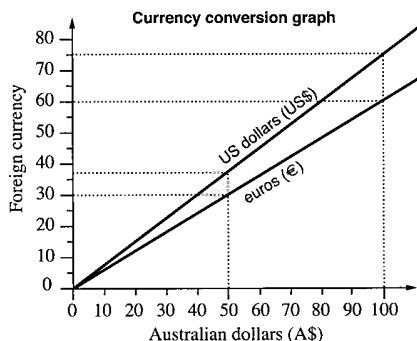
- 16 A factory makes boots and sandals. In any week
- the total number of pairs of boots and sandals that are made is 200
 - the maximum number of pairs of boots made is 120
 - the maximum number of pairs of sandals made is 150.

The factory manager has drawn a graph to show the numbers of pairs of boots (x) and sandals (y) that can be made.



- Find the equation of the line AD . (1 mark) **Medium**
- Explain why this line is only relevant between B and C for this factory. (1 mark) **Medium**
- The profit per week, $\$P$, can be found by using the equation $P = 24x + 15y$. Compare the profits at B and C . (2 marks) **Hard**
(Q24d, 2009 HSC)

- 17 Sandy travels to Europe via the USA. She uses this graph to calculate her currency conversions.



- After leaving the USA she has US\$150 to add to the A\$600 that she plans to spend in Europe. She converts all of her money to euros. How many euros does she have to spend in Europe? (3 marks) **Medium**
- If the value of the euro falls in comparison to the Australian dollar, what will be the effect on the gradient of the line used to convert Australian dollars to euros? (1 mark) **Medium**

(Q24c, 2007 HSC)

- 18 A clubhouse uses four long-life light globes for five hours every night of the year. The purchase price of each light globe is \$6.00 and they each cost \$ d per hour to run.

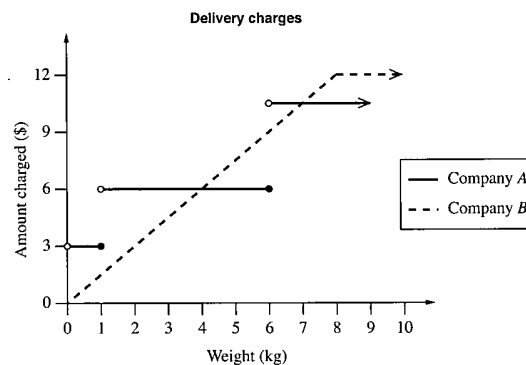
- Write an equation for the total cost (\$ c) of purchasing and running these four light globes for one year in terms of d . (2 marks) **Medium**
 - Find the value of d (correct to three decimal places) if the total cost of running these four light globes for one year is \$250. (1 mark) **Medium**
 - If the use of the light globes increases to ten hours per night every night of the year, does the total cost double? Justify your answer with appropriate calculations. (1 mark) **Hard**
- (Q27b, 2007 HSC)

- 19 Which equation represents the relationship between x and y in this table?

x	0	2	4	6	8
y	1	2	3	4	5

- A** $y = 2x + 1$ **B** $y = 2x - 2$
C $y = \frac{x}{2} - 2$ **D** $y = \frac{x}{2} + 1$ (1 mark)
 (Q7, 2006 HSC) **Easy**

- 20 The graph shows the amounts charged by Company A and Company B to deliver parcels of various weights.



- How much does Company A charge to deliver a 3 kg parcel? (1 mark) **Easy**
- Give an example of the weight of a parcel for which both Company A and Company B charge the same amount. (1 mark) **Easy**
- For what weight(s) is it cheaper to use Company A? (2 marks) **Medium**
- What is the rate per kilogram charged by Company B for parcels up to 8 kg? (1 mark) **Medium**

(Q23d, 2006 HSC)

- 21 A new tunnel is built. When there is no toll to use the tunnel, 6000 vehicles use it each day. For each dollar increase in the toll, 500 fewer vehicles use the tunnel.

- Find the lowest toll for which no vehicles will use the tunnel. (1 mark) **Medium**
- For a toll of \$5.00, how many vehicles use the tunnel each day and what is the total daily income from tolls? (2 marks) **Medium**
- If d (dollars) represents the value of the toll, find an equation for the number of vehicles (v) using the tunnel each day in terms of d . (2 marks) **Hard**
- Anne says 'A higher toll always means a higher total daily income'.

Show that Anne is incorrect and find the maximum daily income from tolls. (Use a table of values, or a graph, or suitable calculations.) (3 marks) **Hard**

(Q28b, 2006 HSC)

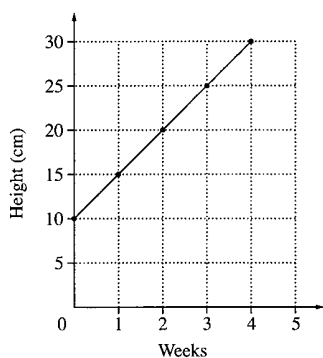
- 22 The total cost, \$ C , of a school excursion is given by $C = 2n + 5$, where n is the number of students.

If three extra students go on the excursion, by how much does the total cost increase?

- A \$6 B \$11
C \$15 D \$16 (1 mark)

(Q17, 2005 HSC) **Medium**

- 23 Susan drew a graph of the height of a plant.



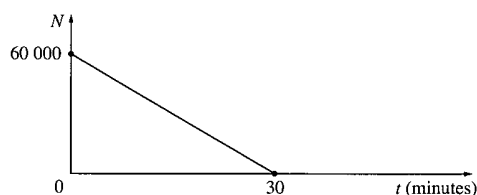
What is the gradient of the line?

- A 1 B 5
C 7.5 D 10 (1 mark)

(Q2, 2004 HSC) **Medium**

- 24 At a World Cup rugby match, the stadium was filled to capacity for the entire game. At the end of the game, people left the stadium at a constant rate.

The graph shows the number of people (N) in the stadium t minutes after the end of the game.



The equation of the line is of the form $N = a - bt$, where a and b are constants.

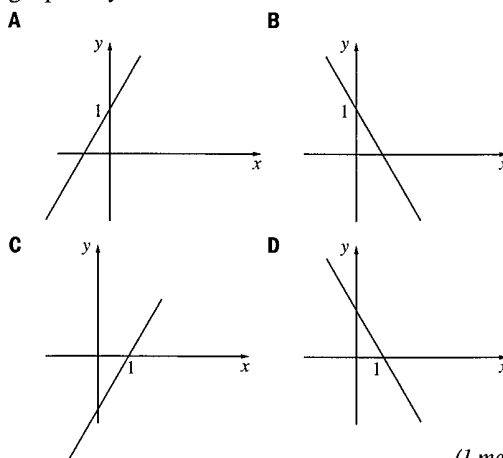
- Write down the value of a , and give an explanation of its meaning. (2 marks) **Medium**
- 1 Calculate the value of b . (1 mark) **Easy**
2 What does the value of b represent in this situation? (1 mark) **Medium**
- Rearrange the formula $N = a - bt$ to make t the subject. (2 marks) **Medium**
- How long did it take 10000 people to leave the stadium? (1 mark) **Medium**
- Copy or trace the graph of N against t shown above into your answer booklet.

Suppose that 15 minutes after the end of the game, several of the exits had been closed, reducing the rate at which people left.

On the same axes, carefully draw another graph of N against t that could represent this new situation. Your new graph should show N from $t = 15$ until all the people had left the stadium. (2 marks) **Hard**

(Q26a, 2003 HSC)

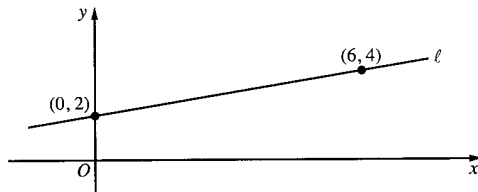
- 25 Which one of the following could be the graph of $y = 3x + 1$?



(1 mark)

(Q6, 2002 HSC) **Easy**

- 26 What is the equation of the line ℓ ?



- A $y = 6x + 2$ B $y = x + 2$
 C $y = 3x + 2$ D $y = \frac{1}{3}x + 2$ (1 mark)

(Q13, 2001 HSC) Medium

- 27 Otto is the manager of a weekend market in which there are 220 stalls for rent. From past experience, Otto knows that if he charges d dollars to rent a stall, then the number of stalls, s , that will be rented is given by:

$$s = 220 - 4d.$$

- i How many stalls will be rented if Otto charges \$7.50 per stall? (1 mark) Easy
 ii Copy and complete the following table for the function $s = 220 - 4d$. (1 mark) Easy

d	10	30	50
s			

- iii Draw a graph of the function $s = 220 - 4d$.

Use your ruler to draw the axes. Label each axis, and mark a scale on each axis. (2 marks) Easy

- iv Does it make sense to use the formula $s = 220 - 4d$ to calculate the number of stalls rented if Otto charges \$60 per stall? Explain your answer. (1 mark) Medium

(Q26a, 2001 HSC)

- 1 $y = -1 - 2x$
If x becomes $x + 5$,
 $y = -1 - 2(x + 5)$
 $= -1 - 2x - 10$
So y decreases by 10.
Answer D

- 2 \$2 per minute = \$120 per hour
So $C = 90 + 120t$
Answer D

- 3 a $m = \frac{4}{7}$
So $p = \frac{4}{7}d$ (1 mark)

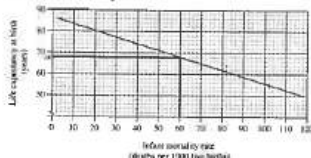
- b $y = 76d$
When $y = 93\,100$
 $76d = 93\,100$
 $d = 1225$ ✓
When $d = 1225$
 $p = \frac{4}{7} \times 1225$
 $= 700$
So 93 100 yen is 700 British pounds. ✓ (2 marks)

- 4 i Rate for company A
 $= \$360 \div 3$
 $= \$120$ per hour (1 mark)

- ii $m = \frac{\text{vertical change in position}}{\text{horizontal change in position}}$
 $= \frac{360 - 75}{3}$
 $= 95$ ✓
 $b = 75$
 $c = mx + b$
 $c = 95x + 75$ ✓ (2 marks)

- iii Cost from company B:
 $c = 95x + 75$
When $x = 5$,
 $c = 95 \times 5 + 75$
 $= 550$ ✓
It costs \$550 to hire the minibus from company B.
Cost from company A
 $= 5 \times \$120 = \600
So it is \$50 cheaper to hire the minibus from company B than from company A. ✓ (2 marks)

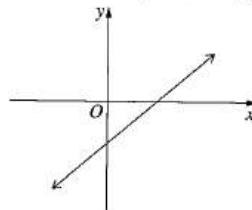
- 5 Life expectancy at birth = 68 years



Answer A

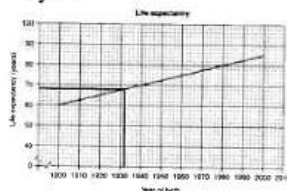
- 6 1 pentagon uses 5 matches.
2 pentagons use 9 matches.
3 pentagons use 13 matches.
Each extra pentagon uses an extra 4 matches.
So the rule for the number of matches is 4 times the number of pentagons plus 1.
Now $100 \div 4 = 25$.
So around 25 pentagons could be formed with 100 matches.
But $4 \times 25 + 1 = 101$ so there are not quite enough matches to make 25 pentagons.
 $4 \times 24 + 1 = 97$
So 24 complete pentagons could be formed.
Answer B

- 7 The gradient, m , is positive.
The y -intercept, b , is negative.



Answer A

- 8 i 68 years

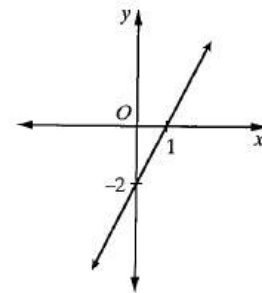


(1 mark)

- ii Gradient $= \frac{20}{80} = \frac{1}{4}$ ✓
The gradient shows the number of extra years of life expectancy for every later year of birth. So for every 4 years later the year of birth is one extra year of life expectancy. ✓ (2 marks)

- 9 gradient $= \frac{-2}{1} = -2$
 y -intercept = 2
Equation is $y = -2x + 2$
Answer A

- 10 $y = 2x - 2$ has y -intercept -2 .
[So the graph could only be A or D.] The gradient is 2.
So the graph can only be D.
[A has a negative gradient.]



Answer D

- 11 Let m be the weight on the moon and e the weight on earth.

Now $m \propto e$

$$m = ke$$

When $e = 84$, $m = 14$

$$14 = k \times 84$$

$$k = \frac{14}{84}$$

$$= \frac{1}{6}$$

$$\text{So } m = \frac{1}{6}e$$

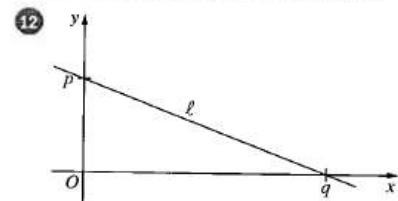
When $m = 2449$,

$$2449 = \frac{1}{6}e$$

$$e = 2449 \times 6$$

$$= 14\,694$$

On Earth, the weight of the landing craft would be 14 694 kg. ✓ (2 marks)



$$\text{gradient} = \frac{\text{vertical change in position}}{\text{horizontal change in position}}$$

$$= \frac{-p}{q}$$

$$= -\frac{p}{q}$$

Answer A

13

Shape (S)	1	2	3
Number of dots (N)	6	8	10

The rule is $N = 2S + 4$

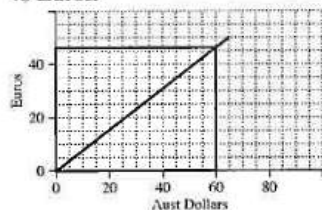
When $S = 156$,

$$N = 2 \times 156 + 4$$

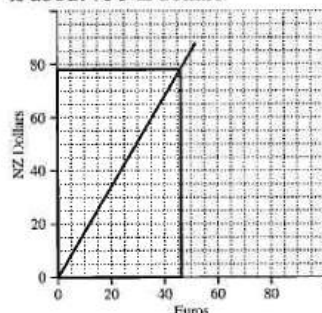
$$= 316$$

Answer A

- 14 From the first graph,
60 Australian dollars is about
46 Euros.

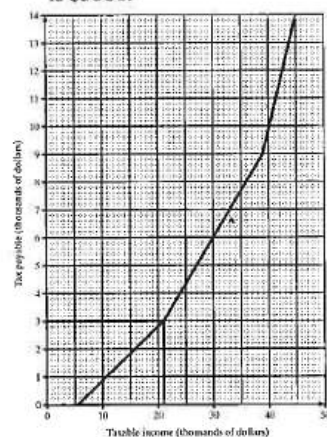


From the second graph 46 Euros
is about 78 NZ dollars.



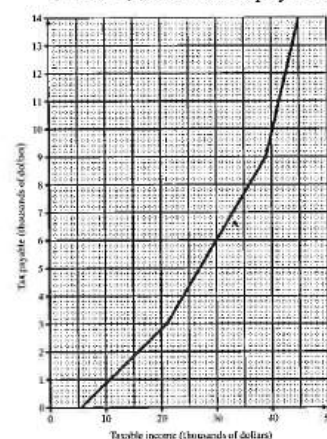
Answer C

- 15 i On \$21 000 the tax payable
is \$3000.



(1 mark)

- ii On a taxable income of
\$39 000, \$9000 tax is payable.



Taxable income (thousands of dollars)

$$\begin{aligned} \text{Gradient} &= \frac{\text{vertical change in position}}{\text{horizontal change in position}} \\ &= \frac{9000 - 3000}{39000 - 21000} \\ &= \frac{6000}{18000} \\ &= \frac{1}{3} \end{aligned}$$

(1 mark)

- iii $\frac{1}{3}$ of each dollar is payable in
tax. So $33\frac{1}{3}$ cents in each dollar
earned between \$21 000 and
\$39 000 is payable in tax.

(1 mark)

- iv The equation is of the form
 $y = mx + b$.

$$\text{Now } m = \frac{1}{3}$$

$$\text{So } T = \frac{1}{3}I + b$$

$$\begin{aligned} \text{Now when } I &= 21\,000, \\ T &= 3000 \end{aligned}$$

$$3000 = \frac{1}{3} \times 21\,000 + b \quad \checkmark$$

$$3000 = 7000 + b$$

$$b = 3000 - 7000$$

$$b = -4000$$

$$T = \frac{1}{3}I - 4000 \quad \checkmark$$

(2 marks)

- 16 i The number of pairs of boots (x)
and the number of pairs of sandals
(y), total 200 in a week.

$$\therefore x + y = 200$$

(1 mark)

- ii $y \leq 150$ as the number of pairs
of sandals must be 150 or less.

$$x \leq 120 \text{ as the number of pairs} \\ \text{of boots must be 120 or less.}$$

(1 mark)

- iii $P = 24x + 15y$

$$\text{At B, } x = 50 \text{ and } y = 150$$

$$\therefore P = 24 \times 50 + 15 \times 150 \\ = \$3450 \quad \checkmark$$

$$\text{At C, } x = 120 \text{ and } y = 80$$

$$\therefore P = 24 \times 120 + 15 \times 80 \\ = \$4080$$

$\therefore$ The profit is greater by \$630
when the number of pairs of
boots made is the maximum
number of pairs of boots that
can be made. $\checkmark$

(2 marks)

- 17 i From the graph,

$$\begin{aligned} \text{US\$75} &= \text{€60} = \text{A\$100} \quad \checkmark \\ \therefore \text{US\$150} &= \text{€120} = \text{A\$200} \\ &\text{and } \text{€360} = \text{A\$600} \quad \checkmark \end{aligned}$$

$$\therefore \text{Sandy will have} \\ \text{€120} + \text{€360} = \text{€480.} \quad \checkmark$$

(3 marks)

- ii If the value of the euro
falls, then A\$100 will buy
more euros than the current
graph shows. Thus the line
will become steeper and the
gradient will be greater.

(1 mark)

- 18 i Cost of purchasing

$$1 \text{ globe} = \$6$$

$$\text{Cost of running}$$

$$1 \text{ globe} = \$d/h \quad \checkmark$$

$$\text{Running cost per year of} \\ 1 \text{ globe}$$

$$= 5 \times \$d \times 365$$

$$= \$1825d$$

$$\therefore \text{Cost of purchasing and} \\ \text{running 4 globes:}$$

$$\$c = 4(\$1825d + \$6). \quad \checkmark$$

(2 marks)

- ii $250 = 4(1825d + 6)$

$$1825d + 6$$

$$= \frac{250}{4}$$

$$1825d = 62.5 - 6$$

$$1825d = 56.5$$

$$d = 0.03095 \dots$$

$$= 0.031$$

to 3 decimal places

i.e. The cost of running each
globe is \$0.031/h.

(1 mark)

- iii The running cost will double
but not the total cost.

$$\text{Running cost per year}$$

$$= 10 \times \$d \times 365$$

$$= \$3650d,$$

but purchase cost of each
globe is still \$6.

(1 mark)

- 19 From the table,
when $x = 0$, $y = 1$
 $\therefore$ A or D is possible.

Test another point, e.g. (4, 3):

$$(A) y = 2x + 1$$

$$\text{When } x = 4, y = 2 \times 4 + 1$$

$$= 9$$

$$\neq 3 \quad \therefore \text{not A.}$$

$$(D) y = \frac{x}{2} + 1$$

$$\text{When } x = 4, y = \frac{4}{2} + 1$$

$$= 3 \quad \therefore \text{D}$$

Answer D

- 20 i \$6 (1 mark)
 ii 4 kg or 7 kg (1 mark)
 iii Weights > 4 kg and ≤ 6 kg, and weights > 7 kg. ✓✓ (2 marks)
 iv From graph, for 8 kg, cost is \$12
 $\therefore$ Cost for 1 kg = $\frac{\$12}{8} = \1.50
 $\therefore$ Rate is \$1.50/kg. (1 mark)

- 21 i Toll (d) No. of vehicles (v)
 $\$0$ 6000
 $\$1$ 5500
 $\$2$ 5000
 $\$3$ 4500
 $\vdots$
 Now $6000 \div 500 = 12$

$\therefore$ When the toll is \$12.00, no vehicles will use the tunnel. (1 mark)

- ii Continuing the table above, when the toll is \$5.00, 3500 vehicles use the tunnel. ✓
 Total daily income = $\$5 \times 3500 = \$17\,500$ ✓ (2 marks)

- iii Number of vehicles = $6000 - 500 \times \text{cost of toll}$
 $\therefore v = 6000 - 500d$ ✓✓ (2 marks)

- iv Total income (I)
 $= v \times d$
 $= (6000 - 500d) \times d$
 $= 6000d - 500d^2$ ✓

Looking at a table of values:

d	0	1	2	3	4
I	0	5500	10 000	13 500	16 000

d	5	6	7	8	10	12
I	17 500	18 000	17 500	16 000	10 000	0

Anne is incorrect as shown by the table of values. ✓
 The maximum daily income is \$18 000 when the toll is \$6.00. As the toll increases above \$6.00, the number of vehicles decreases as does the total income. ✓ (3 marks)

- 22 $C = 2n + 5$
 Let $n = 3$,
 $\therefore C = 2 \times 3 + 5 = \11

If n increases by 3, then $n = 6$.
 $\therefore C = 2 \times 6 + 5 = \17

$\therefore$ Cost increases by $\$17 - \$11 = \$6$.

Answer A

- 23 Gradient = $\frac{\text{difference in height}}{\text{difference in time}}$
 $= \frac{15 - 10}{1 - 0}$
 $= \frac{5}{1}$
 $= 5 \text{ cm/week}$

Answer B

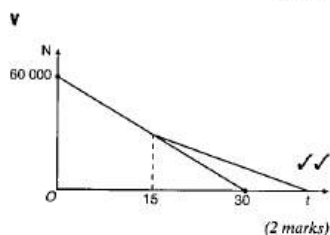
- 24 i $a = 60\,000$ ✓
 This is the number of people who were in the stadium before any started to leave at the end of the game. 60 000 is the capacity of the stadium. ✓ (2 marks)

- ii 1 $b = \frac{\text{rise}}{\text{run}}$
 $= \frac{60\,000 - 0}{30 - 0}$
 $= 2000$ (by calc.) (1 mark)

- 2 b represents the rate at which the people left the stadium, i.e. 2000 people per minute. (1 mark)

- iii $N = a - bt$
 $\therefore N + bt = a$ ✓
 $\therefore bt = a - N$
 $\therefore t = \frac{a - N}{b}$ ✓ (2 marks)

- iv If $a = 60\,000$ and $N = 50\,000$
 $t = \frac{60\,000 - 50\,000}{2000}$
 $= 5 \text{ minutes}$ (1 mark)



- 27 i $s = 220 - 4d$
 $= 220 - 4 \times 7.50$
 $= 190$

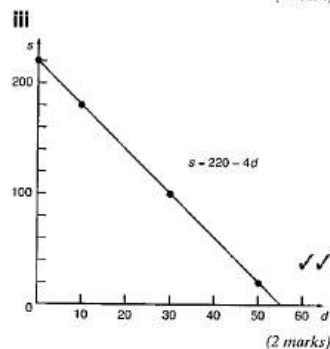
(1 mark)

ii

d	10	30	50
s	180	100	20

When $d = 10$, $s = 220 - 4 \times 10 = 180$ etc.

(1 mark)



- iv No, because $d = 60$ is not a possible value of d as it will give a negative number of stalls. (1 mark)

- 25 $y = 3x + 1$ is in the form $y = mx + b$.
 $\therefore$ The graph passes through 1 on the y axis, and has a positive slope of 3.
 Answer A

- 26 $y = mx + b$
 $m = \frac{4 - 2}{6 - 0} = \frac{1}{3}$
 $b = 2$
 $\therefore y = \frac{1}{3}x + 2$
 Answer D